Myanmar Earthquake & Dam Risk Report

1. Executive Summary

This report covers 9,405 earthquake events in Myanmar from 1970-07-29 to 2026-07-01. The largest event was M7.7 on 1988-11-06. The Gutenberg-Richter b-value is 1.049 with magnitude of completeness Mc=4.7.

Total Events: 9,405

Date Range: 1970-07-29 to 2026-07-01

Max Magnitude: M7.7

Mean Depth: 24.8 km

b-value: 1.049

a-value: 7.715

Mc (completeness): 4.7

2. Earthquake Map

3. Seismic Clustering Analysis

Myanmar Earthquake & Dam Risk Report

4. Dam Seismic Risk Assessment

Risk assessment for 254 dams using PGA estimation from the M7.7 mainshock, fault proximity, and structural exposure scoring.

Critical Risk: 25 dams

High Risk: 148 dams

Moderate Risk: 67 dams

Low Risk: 14 dams

Top 30 Highest Risk Dams:

Name	River	PGA(g)	Dist Fault	Risk	Grade
Dam near 22.09N 95.86E	nan	0.4019	5.7 km	8.3	Critical
Dam near 21.77N 95.89E	nan	0.4626	2.7 km	8.3	Critical
Natkan	nan	0.3593	7.9 km	8.2	Critical
Dam near 21.47N 95.87E	nan	0.3944	6.0 km	8.2	Critical
Ta Nai Hka	Ta Nai Hka	0.4441	3.9 km	8.2	Critical
Dam near 20.72N 95.90E	nan	0.4000	5.6 km	8.2	Critical
Dam near 21.74N 95.85E	nan	0.3656	7.4 km	8.2	Critical
Taungnyo	Taungnyo	0.5047	1.2 km	8.2	Critical
Dam near 20.60N 95.90E	nan	0.3681	7.4 km	8.2	Critical
Dam near 20.47N 96.08E	nan	0.3252	9.9 km	8.2	Critical
Dam near 20.85N 96.03E	nan	0.3478	8.3 km	8.2	Critical
Meikhtila	nan	0.3525	8.2 km	8.2	Critical
Dam near 17.55N 96.37E	nan	0.3298	9.6 km	8.1	Critical
Dam near 21.05N 95.83E	nan	0.3105	11.0 km	8.0	Critical
Dam near 21.04N 95.83E	nan	0.3122	10.9 km	8.0	Critical
Thaphan Chaung	Thaphan Chaung	0.3041	11.5 km	7.8	Critical
Dam near 21.34N 95.81E	nan	0.2956	12.2 km	7.7	Critical
Kun Chaung	Kun Chaung	0.5304	0.9 km	7.6	Critical
Phyu Chaung	Pyu Chaung	0.5211	1.3 km	7.5	Critical
Kabaung	Kabaung Chaung	0.3302	9.2 km	7.5	Critical
Dam near 21.37N 95.80E	nan	0.2830	13.4 km	7.5	Critical
Yenwe	Ye Nwe Chaung	0.4562	3.9 km	7.4	Critical
Nyaunggone	nan	0.2795	13.7 km	7.4	Critical
Dam near 20.73N 95.83E	nan	0.2809	13.6 km	7.4	Critical
Zeedaw	Nyaung Oat	0.2633	15.3 km	7.2	Critical
Male-Nattaung	Male-Nattaung	0.2327	18.6 km	6.9	High
Kyee Ni	nan	0.2362	19.4 km	6.8	High
Kintat	Mu	0.2235	21.0 km	6.7	High
Dam near 20.58N 96.17E	nan	0.2281	20.7 km	6.7	High
Myotha	Ku Htaw	0.2211	20.6 km	6.7	High

5. Aftershock Forecast

Modified Omori law parameters fitted to aftershocks of M7.7 mainshock.

K (productivity): 4.53

c (time offset): 0.100 hours

p (decay exponent): 0.932

Expected M>=3 (7d): 0

Expected M>=3 (30d): 0

Expected M>=3 (90d): 0

Note: These are statistical estimates based on the Omori decay law. Actual aftershock rates may vary due to local geological conditions.

6. Data Sources

- Earthquake data: mmeq.akze.net API (USGS/ISC sourced)
- Dam data: Open Development Mekong (IFC/WLE), CC BY-SA 4.0
- Fault lines: USGS/Plate boundary data
- PGA estimation: Abrahamson & Silva (2008) NGA-West1 GMPE
- Report generated by mmeq-opendata toolkit